



A new dynamic spatial information design framework for AR-HUD to evoke drivers' instinctive responses and improve accident prevention

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ARTICLE INFO

Keywords:

Skills rules and knowledge (SRK) taxonomy
Augmented reality head-up display (AR-HUD)
Situation awareness (SA)
Dynamic spatial information design (DSID)
Dangerous driving scenarios

ABSTRACT

Driver's instinctive responses and skill-based behaviors enable them to react faster and better control their vehicle in dangerous situations. This study incorporated dynamic spatial information design (DSID) in an augmented reality head-up display (AR-HUD) under manual driving conditions. By integrating the skill, rule, and knowledge (SRK) taxonomy and situation awareness (SA) theory, our AR-HUD successfully evoked drivers' instinctive responses and improved driving safety. First, we converted symbol and sign information processed at the knowledge-based and rule-based levels, respectively, into signal information processed at the skill-based level. Then we developed four AR-HUD interfaces with different dynamic designs for use in a hazardous scenario at an intersection. Finally, we investigated each design's impact on drivers' SA and driving performance. Experimental results demonstrated that our DSID enhanced drivers' SA and accident-avoidance capabilities while reducing their cognitive workload. Among the four AR-HUD interfaces, the one that incorporated all three information elements under study (i.e., lateral warning, dynamic driving space, and speedometer) performed the best. This indicates that our proposed framework has potential applications in other similar dangerous driving scenarios, thus contributing to the development of safer and more efficient driving environments.

1. Introduction

Human driving behavior is based on real-time responses to changes in the environment, which involve various levels of cognitive activities. The skills, rules, and knowledge (SRK) taxonomy theory categorizes human cognitive activities into three levels: skill-based, rule-based, and knowledge-based (Rasmussen, 1987). According to this theory, skill-based behavior is instinctive and automated, and humans have the fastest response time to stimuli at this level. The other two levels require the processing of information based on existing rules or knowledge before taking action, resulting in a slower response time. In fact, a significant cause of accidents is the delayed response or lack of perception of what is in the environment by humans. This suggests that incorporating contextual information that triggers faster instinctive responses can reduce the reaction time of drivers and enable them to make faster decisions to avoid danger.

A good option for triggering quick and appropriate instinctive responses and skill-based behaviors in drivers is projecting virtual information into the real environment. The augmented reality head-up display (AR-HUD) is an advanced interface technology that uses AR to overlay information onto the driver's real environment. This creates a synchronized space where real and virtual information can appear in real time together (Moussa et al., 2012).

Using AR-HUDs, drivers can maintain their gaze and attention on the road for longer periods, enabling quicker responses to hazards and thereby enhancing driving performance and reducing workload (Kim and Gabbard, 2022). For example, AR information can be superimposed onto other road users to increase the driver's perception and prediction of pedestrian behavior (Medenica et al., 2011). Additionally, displaying important driving events on the AR-HUD interface can enhance drivers' situation awareness (SA) (Endsley, 1988; George et al., 2012).

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Effective interface design can rapidly direct the driver's attention to critical elements in the environment related to potential hazards, thereby enhancing the driver's SA (Endsley, 1988) and improving driving safety. Unfortunately, while overlaying critical information on an AR-HUD windshield display (WSD) can enhance response times (Yoo, 1999; Horrey et al., 2003), poorly designed projections can lead to various issues and human-machine interaction misunderstandings (Ward et al., 1995). Therefore, many studies have focused on designing rational AR-HUD interface messages to best enhance drivers' SA.

In some studies, SRK taxonomy theory has been used to intervene in the cognitive processes of drivers. For example, Tönnis et al. (2007) designed an AR-HUD interface that displays the estimated braking distance of a vehicle as an AR braking bar on the road ahead. They also mapped the rotation of the steering wheel to the left and right movements of the AR path displayed on the interface. Kim et al. (2013) designed visual information on the AR-HUD that varies according to the distance between the vehicle and the impending danger. In left-turn scenarios, Tran et al. (2013) showed that displaying the projected path of other traffic participants on the AR-HUD can help activate drivers' instinctive reactions and alleviate visual distraction.

The above-mentioned studies are well-targeted at the driver's SA, and use the SRK taxonomy to intervene in interface information design, but they do not specifically address interface design for particular types of dangerous situations. Moreover, they only focus on the impacts of the position and dynamic changes in the semantics of interface elements on drivers. In contrast, we investigate the effects of dynamic changes in interface information content (including color, semantics, and morphology) on the SA and driving performance of drivers. Considering that in the future there is a high probability that a vehicle's whole windshield will be used as an AR-HUD display, all of our AR-HUD interfaces are designed based on the WSD size.

To address the aforementioned gap, the purpose of this study is to explore how to design different levels or types of information in hazardous scenarios to trigger the driver's instinctive response and driving skills as quickly as possible to avoid danger (using the example of hazardous scenarios at intersections). Therefore, our research questions (RQ) are:

- **RQ1:** What SA information in the hazard scenario needs to be shared with the human driver through the interface?
- **RQ2:** What method should be used to design dynamic spatial information that meets drivers' SA needs?
- **RQ3:** What are the impacts of different forms of dynamic information design on drivers' behavior?

To answer **RQ1**, we reviewed SA theory, conducted cognitive work analysis, and improved use-case diagram construction for the driving task. This process enabled us to identify the information that the Advanced Driving Assist System (ADAS) needs to convey to the driver through the AR-HUD. To address **RQ2**, we proposed a dynamic spatial information design (DSID) framework based on the SRK taxonomy and SA theory. We then designed cognitive information for the AR-HUD interface in hazardous intersection scenarios and visualized the dynamic information content of the interface. To answer **RQ3**, we conducted a controlled experiment and obtained quantitative results.

The structure of this paper is as follows: In Section 2, we review the theory and research related to HUDs, SA, and the SRK taxonomy, and present our own conceptual framework. In Section 3, we use the conceptual framework to complete different interface designs. In Sections 4 and 5, we conduct experiments and data analysis. In Section 6, we discuss the results of the experiments. In Section 7, we elaborate on the limitations of this study. Finally, we summarize the work of this study and suggest future work.

2. Context and theoretical framework

In this section, we will review relevant research related to this study, including SA and SRK taxonomy. The SA theory aided us in understanding the information requirements of drivers in the given scenarios, while the SRK taxonomy guided the visualization of information to enhance the driving performance of drivers in hazardous situations.

2.1. AR-HUD related research

Currently, there is a substantial body of research focusing on HUDs. These studies encompass a range of investigations, including comparisons between HUDs and traditional head-down displays (HDDs) concerning their effects on driving capabilities, traffic safety, and driver spatial perception. Additionally, some research has delved into the impact of various types of AR visual elements on a driver's ability to respond to sudden hazards.

Numerous studies have demonstrated that drivers using HUDs maintain higher driving stability and exhibit quicker responses during the driving process than those using HDDs (e.g., Smith et al., 2016). Bram-Larbi et al. (2020) designed an AI-suggested AR path for a HUD interface. They found that compared to traditional in-car dashboards or central screens, HUDs help emergency vehicle drivers maintain their attention more effectively during traffic congestion, enabling them to perform their tasks more efficiently. In a similar vein, Charissis and Papanastasiou (2010) developed collision signs for HUDs for foggy weather scenarios, featuring size and color variations correlated with distance to enhance driver spatial awareness and response speed in low-visibility conditions. They also tested and verified the effectiveness of HUD in delivering critical information during periods of reduced visibility (Charissis and Papanastasiou, 2006; Charissis et al., 2006).

In research examining the impact of various types of AR visual elements on drivers, Kazazi et al. (2015) and others compared the stop sign warning with three other caution sign warnings, demonstrating that different levels of emergency situations should employ distinct warning types. Different warning types can affect a driver's speed control and may interact with the driver's experience in influencing the smoothness of braking (Wu et al., 2018). Chen et al. (2008) and colleagues designed collision prevention features for public buses using AR-HUD, employing icons with different semantics to convey potential collision events and directions. They found that AR visual elements, when accompanied by sound warnings, effectively reduced driver response times to frontal and lateral collision hazards.

In addition to displaying information on AR-HUDs, alternative solutions, such as displaying collision warning patterns in locations such as rear-view mirrors as a substitute for HDD, have been proposed. However, these solutions and design methods lack guidance on how to integrate and express information effectively (Campbell et al., 2007). To enhance effectiveness in scenarios involving pedestrian crossings, Karatas et al. (2020) developed an AR prompt that tracks pedestrians. It was compared to traditional two-dimensional graphical prompts fixed at the bottom of the HUD, demonstrating positive effects on drivers' ability to recognize pedestrian speed and their acceptance and trust in automatic interventions. Meanwhile, Alves et al. (2013) compared three distance warning prompts on AR-HUDs: a triangular outline indicating close distance, an arrow combination with two colors denoting different danger levels, and an animated arrow combination. However, all three prompts were designed in the traditional two-dimensional graphical form fixed on the HUD screen.

2.2. Situation awareness

SA is the ability to understand the current state of the environment in which one is situated (Jones, 2015). From an individual perspective, SA is defined as "the perception of the elements in the environment

within a volume of time and space, the comprehension of their meaning, and the projection of their status in the near future" (Endsley, 1988). Based on this understanding of SA, it can be divided into three levels: perception, comprehension, and projection.

In the face of dangers that appear suddenly or from blind spots in the field of view, or when a driver lacks sufficiently stable visual attention to respond quickly, serious consequences can occur (Smith et al., 2016; Zwahlen et al., 1988). The ADAS, on the other hand, can provide a driver with SA support through an AR-HUD, assisting them in more quickly perceiving potential dangers and preparing for appropriate responses (You et al., 2023). ADAS is being developed and applied to assist drivers in avoiding accidents even when their perception is insufficient or delayed in response. The SA support provided by the system can cover a single level of SA or multiple levels. Therefore, it is extremely important to identify the best ways to encode SA information obtained by the intelligent agent and share it with human drivers. For example, existing in-vehicle human-machine interfaces (HMIs) display warning icons related to different hazardous situations (Nash et al., 2016). However, too many warnings can sometimes increase the driver's cognitive workload or cause distraction.

With increasingly advanced display technology, information being presented to drivers is not limited to simple sounds and images but can also employ animations or AR to enhance human understanding of the scene. One valuable use of AR messages is to direct the attention of the driver. Medenica et al. (2011) added AR warning information on moving vehicles and other road users to enhance visual effects. AR-HUDs have proven to be good media for providing clues to direct the driver's attention to hazardous roadside situations, thus improving driving safety (George et al., 2012; Rusch et al., 2013). These studies made us realize that it is important to emphasize key messages in the interface that are consistent with all phases of driver's SA.

Another important capability of AR-HUDs is conveying predictive information about other traffic participants to drivers. Stange et al. (2022) informed drivers of potential hazards (e.g., pedestrians crossing the road) through an AR-HUD, which increased their trust in autonomous driving systems. Park et al. (2015) presented different information on the AR-HUD to express the level of danger when other vehicles were merging into a lane, and drivers intuitively understood the current situation through AR warning information. AR warning signs can also be used in potential hazard scenarios, such as left-turning vehicle scenarios and predicting collisions at intersections (Calvi et al., 2020; Wu et al., 2019). Kim and Gabbard (2022) investigated the impact of AR elements that gave drivers information on pedestrians crossing the road in different ways on driver distraction. Among them, the virtual shadow element they designed enhanced participants' awareness of pedestrians without affecting their perception of other road elements. Breve and Zhao (2013) also proposed a vision-based reasoning method that displays information corresponding to three levels of SA on the AR-HUD. This method presented different semantic graphics, providing drivers with recommended driving strategies. This method presented different semantic graphics, providing drivers with recommended driving strategies.

Our research determined some of the SA requirements of drivers in hazardous situations through analytical methods. Interface information content and design followed the characteristics of SA at different levels. We used AR-HUD interface design to deliver information to drivers that is easy to perceive, understand, and predict, thus meeting their SA requirements.

2.3. Skills, Rules, and Knowledge taxonomy (SRK)

After receiving SA information, the driver will provide feedback such as driving decisions based on environmental changes and important information to ensure safe vehicle operation. This cognitive activity is hierarchical and is described by the SRK taxonomy proposed

by Danish psychologist (Rasmussen, 1983), which has been widely applied in fields such as aviation and HMI design (Miller, 1988, 1991).

The SRK taxonomy separates human cognitive activity into three levels: skill-based, rule-based, and knowledge-based (Rasmussen, 1983, 1987). These three levels correspond to different cognitive processes.

2.3.1. Skill-based behavior

Skill-based behavior is a smooth, automatic, and highly integrated behavioral pattern that is triggered by the driver's intention state when facing familiar events (Rasmussen, 1987). It refers to skills that can be improved through practice and repeated use. The execution of these skills is automatic and does not require in-depth thinking or decision-making; rather, a certain amount of learning and training is required before mastery can be achieved. For example, skill-based behavior in driving is highly automated (Vicente and Rasmussen, 1988) and represents the driver's instinctual perception of the vehicle's motion. When drivers use skill-based behavior for activities, they follow instincts to handle constraints in the road environment and relationships with other traffic participants, such as emergency braking when faced with imminent danger. This type of behavior has a low cognitive load.

2.3.2. Rule-based behavior

Rule-based behavior refers to cognitive activities that can be described by simple rules (Schewe and Vollrath, 2021). These rules can come from everyday experience, professional knowledge, or some kind of norm. At this level, people need to do some thinking and decision-making, but these decisions are usually based on simple rules. For example, stopping at a traffic signal based on the color of the traffic light, etc. (Rasmussen, 1983). Rule-based behavior is a non-automated behavior. First, the driver recognizes the traffic signal, then links it to the driving task state, and finally executes the action based on the known traffic rules. When drivers use rule-based behavior for activities, they need to bear the cognitive load of recognizing and reacting to signals, such as adapting the vehicle speed to the road speed limit (Donges, 1999), by comparing the speedometer display information with the speed limit sign that represents the road rule.

2.3.3. Knowledge-based behavior

Knowledge-based behavior refers to cognitive activities that require in-depth thinking and analysis based on existing knowledge reserves. These activities often involve multiple factors. At this level, people need to integrate different information and knowledge, conduct comprehensive analysis and evaluation, and make final decisions. For example, in driving, knowledge-based behavior involves the driver specifying clear goals (such as the driving route) based on the environment and overall analysis, processing symbols in the environment, understanding and analyzing changes in the situation during task execution, and finally making decisions. Knowledge-based behaviors are thus non-automated behaviors that are triggered when drivers face unfamiliar and non-normal events, and as such have a high cognitive load (Cohen et al., 1990; Miller and Cohen, 2001; Shiffrin and Schneider, 1977).

From the above, it is clear that cognitive activities at different levels correspond to different reaction times. Cognitive activities (and thus reaction times) are also related to the information received from the environment. Therefore, we believe that the interface information of AR-HUDs can be designed to encourage drivers to use skill-based behaviors or rule-based behaviors, thereby accelerating their response time in dangerous situations and improving the efficiency and quality of their cognitive activities.

In design research, there is a concept referred to as the ecological interface, which is also developed based on the SRK taxonomy (Hajdukiewicz and Burns, 2004). AR-HUD information designed with an ecological interface can provide drivers with directly perceivable and easy-to-understand information about the environment and other traffic participants. For example, in a study by Ng-Thow-Hing et al. (2013), to assist drivers in safe left turns, the future path of oncoming vehicles

was displayed on the AR-HUD with shape changes that vary with speed. They found that users' visual attention was primarily focused on the movement of oncoming vehicles and paid little attention to the projected path. In another interface design, they also created a grid box surrounding the ego vehicle on the AR-HUD. The color and transparency of this grid box vary with distance, which to some extent caused drivers to stare continuously. They found that drivers made more frequent lane changes in response to this information, which somewhat inhibited their accurate perception of the situation. The graphical changes in AR can directly affect drivers' perception of behavior, stimulating drivers' unconscious, low-workload perception-motion coupling behavior (Schewe and Vollrath, 2020, 2021). For example, a speedometer that changes length with the driver's throttle or brake actions (Schewe and Vollrath, 2020, 2021) belongs to a signal that represents spatiotemporal variables. Schall and Gavin (2010) found that using static danger prompts increased drivers' reaction times compared to no prompts, while dynamic danger-related clues reduced drivers' reaction times compared to static danger prompts. The variations in the display elements in existing ecological interface designs are solely associated with the driver's driving behavior. This can occasionally lead to drivers executing incorrect actions to achieve the desired visual effects (Merenda et al., 2018). Furthermore, most of the visualized information has only one meaning.

As with the three types of behaviors mentioned above, studies have shown that dynamic information cues are beneficial for activating drivers' skill-based behavior. However, the information design in these studies focuses only on variations of a single element, such as shape, color, or semantic variations of a graphic, and does not delve into the question of whether other types of visual information are needed in conjunction with the different phases of a complete event. In our study, we integrated SRK taxonomy and SA to design different information that dynamically changes in shape, color, and semantics for dangerous traffic scenarios.

3. Designing process and conceptual framework

Based on the SRK taxonomy and SA, we proposed an AR-HUD interface design framework and outline the design steps. We will then describe the specific interface design for a given scenario based on this framework and steps. As shown in Fig. 1, the interface design process studied in this paper is based on SA and the SRK taxonomy, which established a framework for designing dynamic spatial information. This was followed by an analysis of cognitive work in hazard scenarios, with adapted use-case diagrams and goal-directed task analysis (GDTA) (Endsley, 2001) to derive SA requirements and specific AR information content for human drivers. Then, the AR information content was visualized for specific scenarios, resulting in an AR-HUD interface visualization form. Finally, the interface design was evaluated through user testing, and based on user feedback, the interface design was iterated to produce the final AR-HUD interface.

3.1. SA requirements analysis

To answer RQ1, we first needed to analyze the SA information involved in the scene. GDTA is commonly used to determine which aspects are important for drivers' SA in a given scenario (Endsley, 2001). It identifies the SA requirements of decision goals and sub-goals, indicating not only what information the system needs to acquire but also how to combine this information to address these goals. The SA requirements correspond to the dynamic information related to the goals during the task completion process. This dynamic information differs from static information such as knowledge and rules (Endsley, 2001). Endsley and Rodgers (1994) gave an example that showed the SA requirements analysis for an air traffic controller for the sub-goal "Maintain Aircraft Conformance" of the major goal "Avoid Conflicts". GDTA systematically defines the requirements at three levels of

SA and determines how various dynamic information can be combined to form the content that drivers truly want to know. Furthermore, GDTA specifically focuses on dynamic situational information that influences drivers' behaviors (Endsley, 2001). For drivers, all objects in the traffic environment are considered dynamic situational information, including pedestrians. The three levels of SA refer to the perception of the elements in the environment (level 1), comprehension of the current situation (level 2), and projection of future status (level 3). Therefore, we conducted a GDTA to analyze the goals and tasks involved in the scenario, and to determine the driver's cognitive information requirements (Endsley, 2001).

As shown in Fig. 2, in the dangerous driving scenario, the driver's overall goal is safe driving. Under the overall goal, there are three sub-goals, namely: monitoring vehicle status, controlling the vehicle, and observing road conditions. Under the overall goal, there are three sub-goals: monitoring vehicle status, controlling the vehicle, and observing road conditions. Under the first sub-goal, the driver's SA requirements are current position (SA level 1), current speed (SA level 1), environmental limitations (SA level 2), speed limitations, and displacement trend (SA level 3). Under the second sub-goal, the driver's SA requirements are potential hazards (SA level 1), other traffic participants' intentions (SA level 2), and the impact of actions (SA level 3). In a study conducted by Merenda et al. (2018), different driving tasks were categorized as goal-directed tasks and stimulus-driven tasks. Events like pedestrians suddenly crossing the driving path require drivers to engage in stimulus-driven tasks, where they cannot anticipate what will happen and can only react based on the changing situation (Merenda et al., 2018). The third sub-goal, observing road conditions, can be considered as a task objective within stimulus-driven tasks. To accomplish such a task objective, drivers typically employ a bottom-up processing information pattern. Under this sub-goal, the driver's SA requirements are safety status (SA level 1), danger level (SA level 2), and changes in condition (SA level 3).

3.2. Identify SA elements

After identifying the SA information required by human drivers, we need to deduce the related SA information that can be provided by the ADAS. In this paper, ADAS specifically refers to an information visualization system that helps the driver to perceive hazards as well as better avoid them. This system provides visual information assistance to the driver via an AR-HUD (WSD) under manual driving conditions. The information that drivers need can be obtained through a use case diagram (Jacobson, 1993), which can be used to establish the relationship between different system functions and the different task goals of drivers (Marshall, 1992). As shown in Fig. 3, in a manual driving scenario, the driver has specific driving goals and corresponding SA requirements. The ADAS provides information to the driver through its various information modules to support the driver's SA.

The abstract goals, tasks, and functions of both the driver and the vehicle system are:

- Human: The overall goal is safe driving, which can be divided into three sub-goals: vehicle status monitoring, observation of road conditions, and vehicle control. To achieve these goals, the driver must perform tasks such as speed control, hazard perception, and hazard assessment when facing sudden dangerous situations. Real-time information such as vehicle speed, road speed limit, road conditions, and the behaviors of other traffic participants is required by the driver.
- ADAS: Includes a lane departure warning system, collision avoidance system, and/or pre-collision system. These functionalities are supported by in-vehicle end-user computing units, along with sensors such as millimeter-wave radar and single or dual cameras serving as video sensors. The detection range of these sensors spans 30–150 m. When the vehicle is traveling at speeds ranging

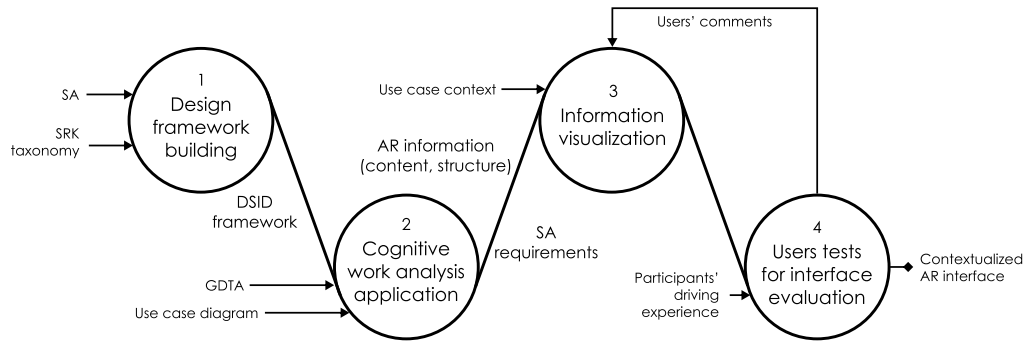


Fig. 1. Our design process for the dynamic spatial interface. Note that the first step is to establish a framework of DSID, which is followed by analyzing cognitive tasks in hazardous scenarios by use case diagrams and GDTA (Endsley, 2001). These analyses help to derive SA needs and specific AR information content. Then, the AR information content is visualized for specific scenarios, resulting in AR-HUD interface visualization forms. Finally, the AR-HUD interface design is evaluated through user testing, and based on user feedback, the interface design is iterated to produce the best version.

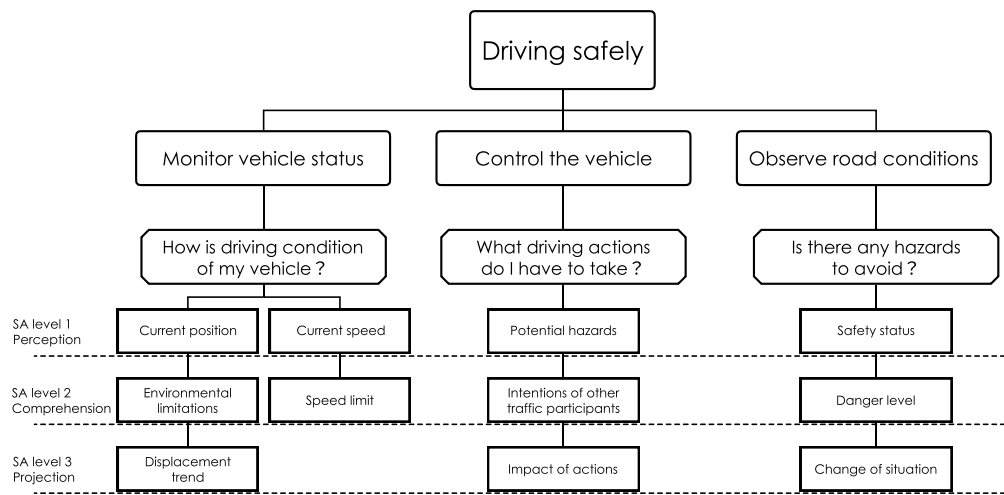


Fig. 2. Goal directed task analysis of hazard scenarios to determine important aspects for drivers' SA. Note that GDTA systematically defines the requirements at three levels of SA (perception of environment, comprehension of current situation, and projection of future status) and determines how various dynamic information can be combined to form the content that drivers truly want and need to know. Furthermore, GDTA specifically focuses on dynamic situational information that influences drivers' behaviors (Endsley, 2001).

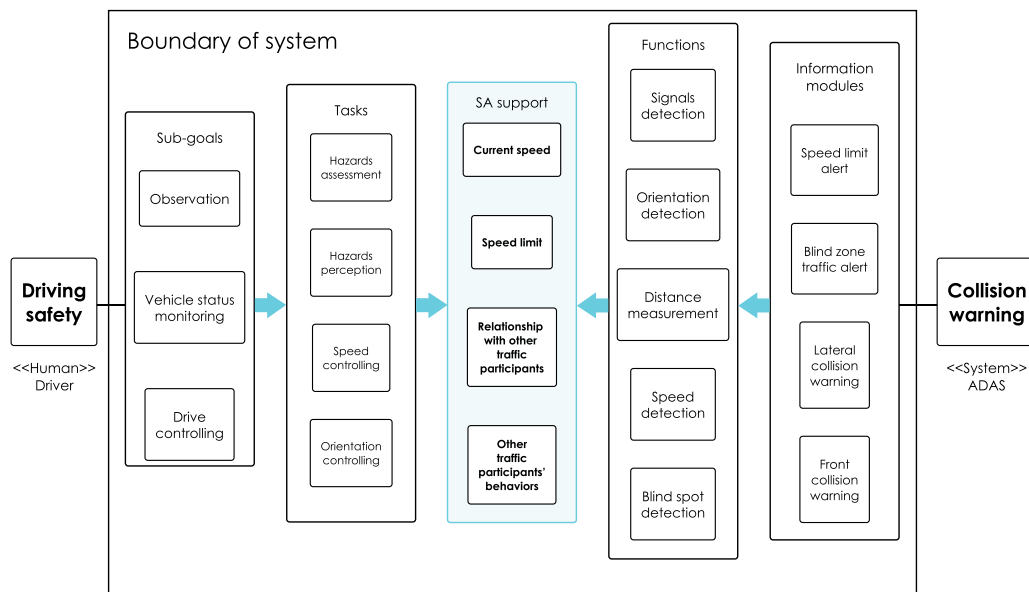


Fig. 3. Adapted use-case diagram for driving task. The white boxes represent the goals and tasks of the human and information modules and the functions of the collision warning system. The blue-filled areas are the information that can be provided by the ADAS, which also represents the AR information content.

from 20–70 km/h, with no turn signals activated, a steering wheel rotation angle of less than 5 degrees, and hazard lights off, the system conducts an analysis considering the distance to other vehicles and relative speeds. Based on this analysis, when a collision risk is identified, it presents visual warning alerts via an AR-HUD (WSD). Consequently, in this scenario, the primary objective of the ADAS is collision prevention. To enable the driver to manage vehicle speed effectively, discern the intentions of other traffic participants, and avoid hazards, support from information modules like speed limit warning, blind spot detection, side collision warning, and front collision warning is indispensable. These information modules provide functionalities such as signal detection, orientation detection, distance measurement, speed detection, and blind spot detection. These functions collectively provide information to the driver, enhancing the driver's SA.

Through the adapted use-case diagram, we can identify that there are four pieces of information that collision warning systems can provide to drivers: speed limit, current speed, relationship with other traffic participants, and other traffic participants' behaviors. These four pieces of information meet the shared SA requirements of the human-vehicle team during general driving and even in dangerous situations.

3.3. Dynamic Spatial Information Design (DSID)

To answer **RQ2**, we needed to convert the information obtained from Sections 3.1 and 3.2 into dynamic spatial information. To achieve this, we needed to understand the types of information that trigger different levels of driver behaviors and match the interface information content accordingly. According to the SRK taxonomy, there are three types of information used to trigger human skill-based, rule-based, and knowledge-based behaviors: signal, sign, and symbol, respectively. The definitions of these three types of information are shown in Table 1.

In hazardous driving scenarios, accidents occur when drivers fail to manage their relationship with other traffic participants and have insufficient time to complete the cognitive process of understanding symbols. Based on the above three types of information, the most important information in dangerous traffic situations should be dynamic spatial information, which needs to be designed in terms of semantics, shape, and color so that the information changes continuously in time and space, and the driver's skill-based behavior can be mapped to the continuous changes in the information.

Fig. 4 illustrates the transformation of the three types of information in our DSID framework, taking the design of lane departure warning as an example. The ADAS obtains situational perception elements from the environment and visualizes them, which are transmitted to the driver through the AR-HUD. The information transmitted to the driver by AR-HUD may be symbol, sign, or signal.

Symbol-type information often appears in conventional driver vehicle interfaces and can be designed as either screen-fixed or world-fixed representations (Gabbard et al., 2014). In the example of a lane departure warning, if the system only displays a red lane and vehicle combination icon, this type of visual information falls under the category of symbol. This type of information requires the driver to rely on their prior life experience and knowledge to infer its meaning of lane departure. At this point, the driver also needs to perceive the specific situation in the actual environment (i.e., which direction the vehicle is departing from the lane) and then adjust the vehicle's driving state to eliminate the warning. If multiple attempts fail to eliminate the warning (i.e., change the visual information being provided on the AR-HUD), the driver will be confused and suspect that the system function has failed. This could lead to drivers ignoring such warnings in the future.

If the system displays an icon representing the warning that the vehicle is deviating from the lane to the left, this visual information is

classified as a sign. At this time, the driver steers the vehicle to the right based on the meaning expressed by the sign and restores the appropriate position. Nowadays, the vast majority of lane departure warnings are designed as animated signs; that is, the lateral displacement of the vehicle corresponds to the movement of the vehicle in the visualization. Sign-type information not only appears on the current conventional driver-vehicle interface using screen-fixed and screen-animated representation methods but will also be present on future AR-HUDs using the world-fixed representation method after undergoing conformal design approaches (Merenda et al., 2018).

However, the driver still needs to go through the coordinate system conversion from the graphical interface to the real space when dealing with this type of sign warning, which is why signal warnings are often preferable. The nature of signal-type information itself determines that it is best represented using the world-fixed or world-animated methods (Merenda et al., 2018). If the lane departure warning is designed as dynamic spatial information, it can directly help the driver perform an appropriate skill-based behavior. The AR dynamic driving area superimposed on the road will change with the control of the steering wheel by the driver. The lane departure situation will be intuitively displayed in the real driving space. Then, the driver would not need to perform additional spatial coordinate system conversion; they would only need to adjust the steering wheel to keep the AR dynamic driving space in the lane of the real environment to complete the task of remaining in their lane.

In complex traffic environments, not all pieces of information are amenable to a single design approach for expression. Therefore, AR-HUDs (WSD) provide visual information using various design methods. For instance, they may superimpose path guidance on the actual road, utilizing either the world-fixed or world-animated approach, while simultaneously presenting the current road speed limit value in a screen-animated format. The interface elements from each design approach can be meaningfully combined and presented on the AR-HUD (WSD) (Merenda et al., 2018). Therefore, we do not intend to confine ourselves to a specific interface design approach or to a particular level of information type within the SRK taxonomy in our design.

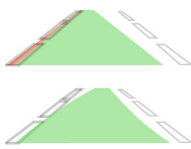
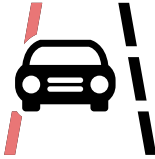
3.4. Designing HMI

3.4.1. Scenario and cognitive task analysis

In this study, we selected the dangerous scenario (see Fig. 5) where the ego car and the right-turning vehicle encounter at an intersection without traffic lights. In fact, intersection-related accidents account for about 20%–45% of all reported traffic accidents in developed and developing countries (Ma et al., 2021; Golembiewski et al., 2011). In all intersections (including uncontrolled intersections) of China, vehicles making a right turn followed by a straight movement are supposed to yield to vehicles traveling straight through the intersection. In the research scenario of this study, according to the rules, the participants' vehicles were consistently traveling straight, so they were not required to yield. However, non-compliance with traffic rules by other road users often leads to highly dangerous situations. Without the guidance of traffic lights, yielding between vehicles relies solely on understanding other vehicles' intentions. However, this is a higher-level cognitive behavior based on knowledge and rules, and drivers must pay a high cognitive load in predicting other vehicles' intention-related cues (Cooper et al., 2003). At this point, drivers may have difficulty processing the information in the scene in a timely manner based on the knowledge and rule hierarchy, leading to an inability to use appropriate skill-based behaviors to avoid danger, which may result in serious accidents (Engström and Markkula, 2007).

To effectively apply the SRK taxonomy to the design of the AR-HUD interface, we not only consider the human user's mental model, but also the machine's information acquisition and processing patterns. We adopted the theory proposed by Parasuraman et al. (2000), which

Table 1
Comparison of three information types.
Source: Adapted from Rasmussen (1983).

	Signal	Sign	Symbol
Definition	Sensory data which directly represents time-space variables from the dynamic spatial configuration in the environment. Boundaries of acceptable performance are directly perceivable.	The state of the environment according to some behavioral rules. They are related to certain rules and action conditions in the environment. Signs are used to activate stored behavior patterns, such as speed limit warnings that activate the driver's deceleration behavior.	Various possibilities, relations, and attributes of changes. They are related to the characteristics of the dynamic environment through certain rules and conventions of events and their relationships and processes.
Advantages	Intuitively explain the environmental state. It is easy for users to respond to most situations directly with their skilled abilities, requiring minimal cognitive resources.	Easy to activate familiar behaviors for operations adjustments.	Can explain unfamiliar events in the environment.
Disadvantages	The often displayed graphical information may cause attention tunneling.	Cannot be used for functional reasoning, generating new rules or predicting the effects of unfamiliar events.	The meaning is defined by the expression of internal concepts and requires reasoning.
Examples	 The graphic, positioned in the real-world environment, dynamically presents the future tendency of the lane deviating to the left. The driver can directly adjust the steering to eliminate the deviation trend.	 The combination of vehicle and lane graphics represents the left lane departure, which requires the driver to compare this state with the actual situation and make adjustments to their driving behavior accordingly.	 Composed of vehicle and lane graphics, this combination signifies that the current vehicle is not in the correct lane position. The driver needs to interpret the graphics based on existing knowledge or experience, understand the real-world driving situation, and subsequently adjust their driving behavior.

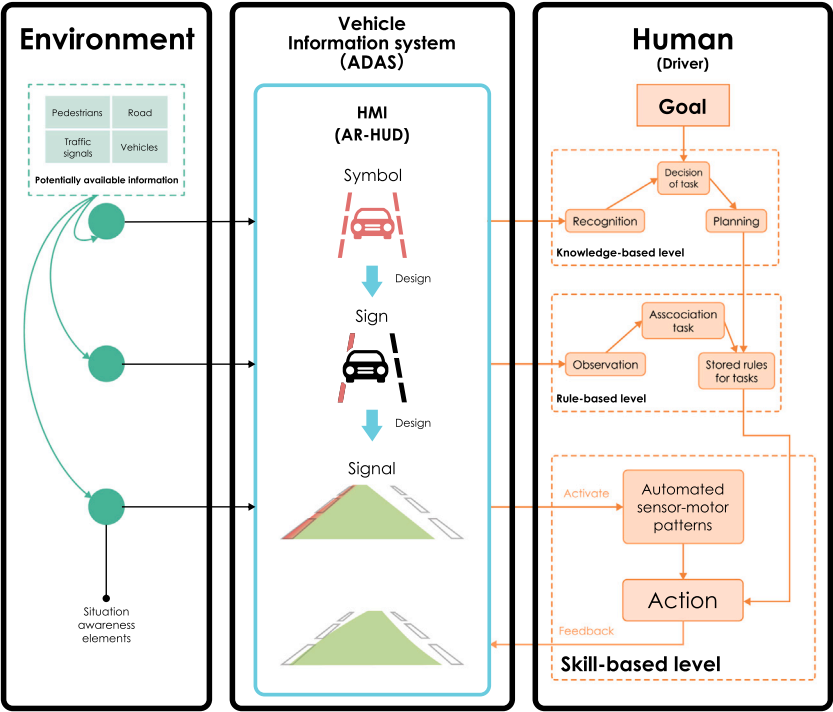


Fig. 4. Our DSID framework. Taking a lane departure warning as an example, we demonstrate three different information designs that can be displayed by an HMI for the same hazardous event. The semantics of symbol-type information require drivers to reason based on existing knowledge combined with environmental status, going through knowledge-based and rule-based levels before activating skill-based behaviors, causing a high cognitive load for drivers. The semantics of sign-type information can be obtained from the graphics themselves combined with existing environmental features and behavior rules, resulting in a lower cognitive load for drivers than experienced when presented with symbol-type information. By designing information as signals, the boundaries of acceptable performance are directly perceivable, and drivers can act with the lowest cognitive load while receiving feedback from the dynamic graphics' changes in the real environment.

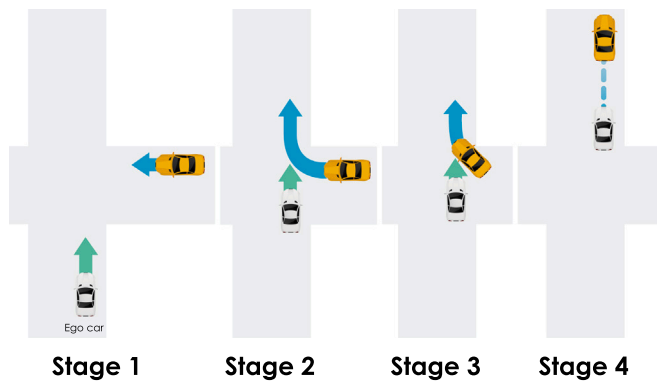


Fig. 5. Four stages of avoiding intersection side hazards. Note that an Ego car that can successfully avoid a collision with another vehicle coming from the side will go through the four stages depicted in the figure, and the shortening of the length of the arrow in the figure represents the deceleration of the Ego car.

identifies four functions of automation: information acquisition, information analysis, decision-making, and action execution. Based on the level of automation in manual driving, we adjusted these automation functions and mapped them to the cognitive and behavioral stages of human drivers. We analyzed the driving tasks in the hazardous scenario involving the Ego car and the turning vehicle at an intersection without traffic lights using the analytical method mentioned in Sections 3.1 and 3.2, as shown in Fig. 6.

We present below the details of the information processing stages in which human drivers and vehicle information systems are involved during driving tasks.

- Driver:

State 1: In the perception stage (SA level 1), the SA demands are current speed, safety status, and potential hazards.

State 2: In the comprehension stage (SA level 2), the SA demands are speed limit, danger level, and other vehicles' intentions.

State 3: In the projection stage (SA level 3), the SA demands are situation changes and the impacts of actions.

State 4: In the decision-making stage, the driver takes the action of decelerating to avoid danger.

- ADAS:

State 1: In the information acquisition stage, speed detection, blind spot detection, and collision detection are performed, which can provide easily perceivable information to the driver (easily perceivable information may be prominent warning).

State 2: In the information analysis stage, analysis of speed limits, collision possibilities, and other traffic participants' behaviors is provided to the driver in a way that is easy to understand and helps them to predict changes in the situation.

State 3: In the feedback stage, the current dynamic dangerous situation is visualized through AR-HUD.

3.4.2. Design schemes

Based on the design framework introduced in Section 3.3 and the analysis of cognitive tasks in the aforementioned scenario, the next step is to design an information display for our AR-HUD interface. According to the driver's information needs, the system must recognize and display the situation of the road ahead while driving forward.

In general, our approach to designing the AR-HUD interface with the DSID framework begins with a flat presentation, then progresses to a semi-spatial presentation, and finally culminates in spatializing the information.

- Flat presentation: In this stage, all driver information requirements and corresponding functional information are presented in a traditional two-dimensional format on the interface. At this point, all the information is temporarily represented in a screen-fixed or screen-animated format. ADAS must provide both lane- and speed-related cues to the driver while conveying dynamic hazard situations and early warning information.
- Semi-spatial presentation: Since drivers have a strong demand to view early warnings, speed, and dynamic hazard information, and they need to slow down, prioritizing the spatial representation optimization of these pieces of information becomes crucial. This stage also involves transitioning information from the knowledge-based level to the other two levels. Therefore, the design strategy at this stage is to position warning information in the direction of the impending danger and represent the deceleration effect and dynamic hazard changes on the forward road. The remaining information can retain the forms used in the previous stage.
- Information spatialization: Designers need to integrate and optimize the semi-spatial information to avoid redundancy. For instance, warnings can appear well in advance when approaching an intersection and disappear when dynamic hazard situations emerge. Dynamic hazard situations and deceleration effects should be intuitively represented on the forward road. Combining conclusions or design recommendations from existing studies, such as the effectiveness of world-animated forms in capturing attention compared to two fixed methods, or the use of animations independent of vehicle motion to convey real-world dangers, and the rapid attention-guiding ability of fixed-world conformal graphics (Merenda et al., 2018), designers can then employ dynamic changes in the shape of the driving space on the road to convey deceleration effects, as well as color and coverage area changes to represent dynamic hazard situations. This completes the process of translating key information into the skill-based level.

The dynamic driving space on the road ahead is often displayed in green, indicating a clear and safe road. When a nearby vehicle is detected that may cause danger, the AR-HUD will display a yellow triangular warning symbol to guide the driver's attention to the dangerous area. We then correspond the change in the distance between the cars in the danger situation with the change in the area of the red zone, where red is also the semantic for danger. Such a change is consistent with the definition that dynamic spatial information changes continuously in space and time with the environment. To map the driver's skill-based behavior to continuous changes in the spatial and temporal attributes of the information, we make the length of the dynamic driving area vary with the degree of braking by the driver. At the same time, the length of the dynamic driving space also represents the expected braking distance of the vehicle. In this way, the driver can intuitively understand the changes in the distance between vehicles, the degree of danger, and the impact of speed control behavior in the current driving environment from the representation of the dynamic driving space.

Fig. 7 depicts four AR-HUD (WSD) interfaces, HMI1–HMI4. Both HMI1 and HMI2 employ the DSID for the AR dynamic driving space. The dynamic driving space visualizes the real-time situation of the road ahead, showing green when the road is safe and gradually being covered in red as danger approaches. The area and rate of change of the red region correspond to the level of danger. The longitudinal length of the dynamic driving space changes as the driver slows down the vehicle, which reflects the expected braking distance of the vehicle. The visual mapping of such information is aimed at activating the driver's behavior pattern based on skill hierarchy, helping them use automatic skills to avoid danger quickly and appropriately. HMI2 has an additional feature of providing early warnings of oncoming side vehicles before entering the intersection, 2.5 s in advance (Srinivasa, 2002; Nakaoka et al., 2008), compared to HMI1.

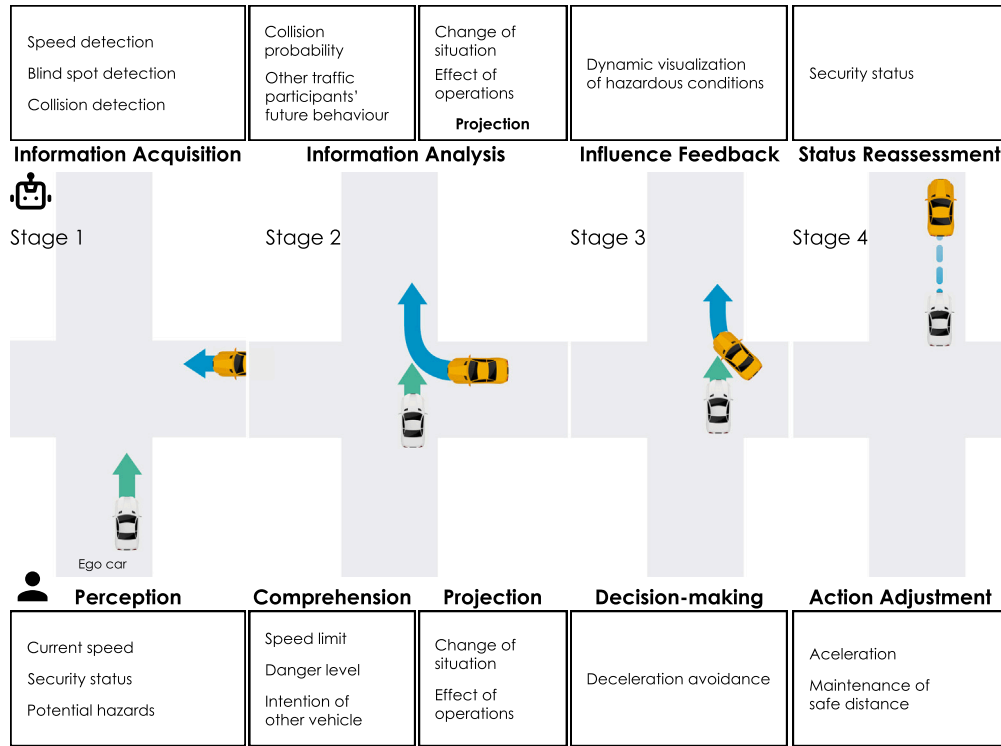


Fig. 6. Intersection scenario and cognitive task analysis of this scenario. Human drivers in the scenario studied in this research go through stages corresponding to the three levels of SA, as well as the subsequent decision-making and action adjustment stages. The AR-HUD system also goes through several stages, including information acquisition, analysis, influence feedback, and status reassessment. The information analysis stage specifically supports the driver's comprehension and projection stages in SA. The information design of an AR-HUD needs to meet the driver's information requirements.

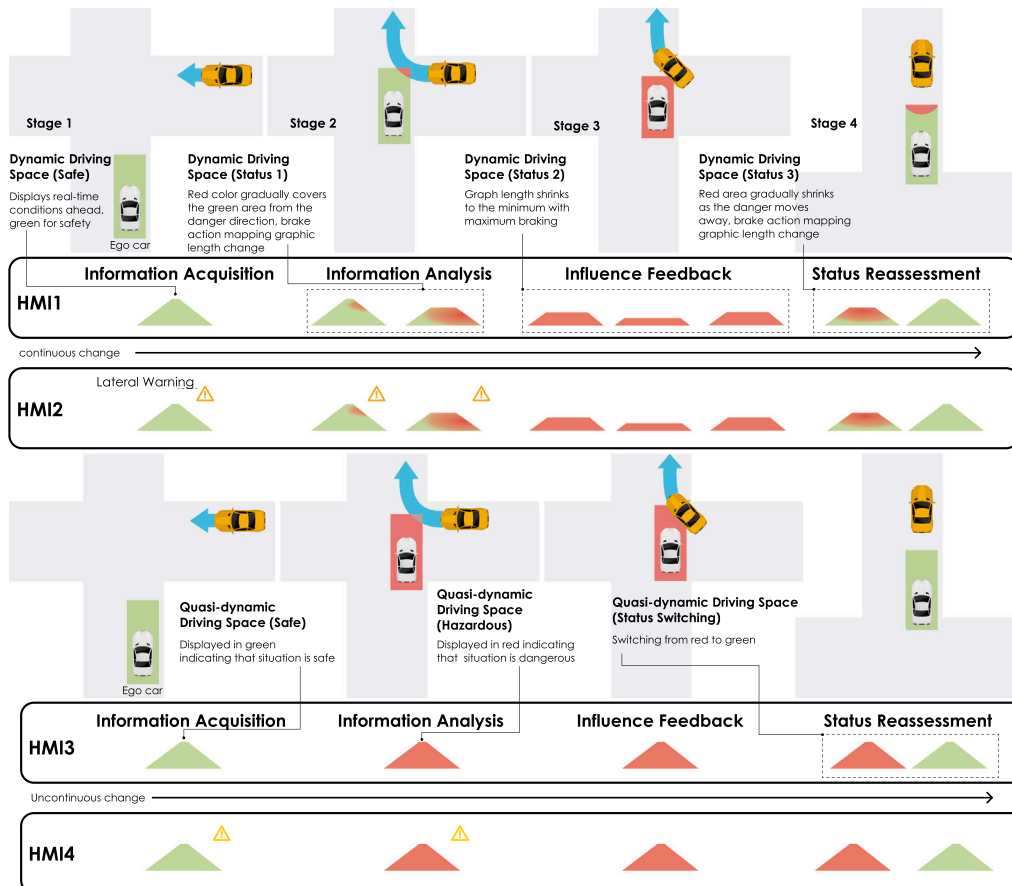


Fig. 7. Designs of HMI1, HMI2, HMI3 and HMI4. HMI1 and HMI3 do not provide any alerts during the information acquisition phase. HMI2 and HMI 4 do provide an alert in the information acquisition phase. Note that HMI1 and HMI2 adopt DSID, while HMI3 and HMI4 use quasi-DSID.



Fig. 8. Areas of interest (AOI). We established two AOIs, one in front of the Ego car in the lane ahead, the other to the side where the lateral warning is located.

HMI3 and HMI4 both utilize a quasi-DSID. The quasi-dynamic driving space is typically displayed in green in a safe state. When the danger approaches to a certain degree, the quasi-dynamic driving space changes from green to red. As the situation changes and the driver moves away from the dangerous condition, the status of the quasi-dynamic driving space switches back from red to green. Besides the common quasi-dynamic driving space element, HMI4 also has an advance warning of lateral traffic 2.5 s before entering a crossroads (Srinivasa, 2002; Nakaoka et al., 2008), while HMI3 does not.

To provide a clear and concise description of the characteristics of the four design schemes, we will refer to the dynamic driving space in HMI1 and HMI2 as Dynamic Spatial Information (DSI), and the quasi-dynamic driving space in HMI3 and HMI4 as Quasi-Dynamic Spatial Information (QDSI). The lateral warning will be referred to as LW, and these designations will be indicated in parentheses when mentioning the four design schemes.

4. Methods

4.1. Experimental design

4.1.1. Hypotheses

The purpose of this research is to explore the impacts of the four different forms of dynamic information design mentioned in Section 3.4.2 on driver SA and performance in dangerous situations (RQ3). We propose the following hypotheses:

H1: DSID can help enhance drivers' SA and improve driving performance.

H2: DSID can reduce drivers' reaction times.

4.1.2. Independent variables

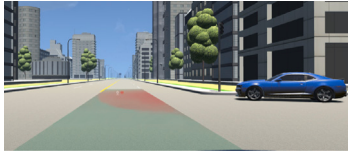
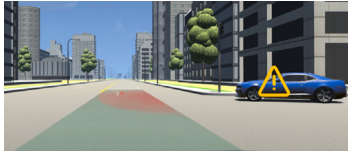
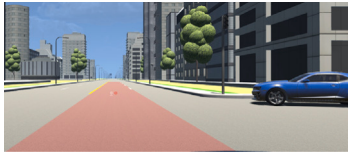
We used a paired repeated measures experimental design. The experiment's independent variables were the four AR-HUD interface designs. Table 2 shows in detail the differences in information types between the four schemes.

4.1.3. Dependent variables and measurements

The dependent variables examined in this experiment were eye-tracking data, driving performance, usability, cognitive workload, and SA, in consideration of the two hypotheses.

- 1. Average fixation duration, number of fixations, average visit duration, and number of visits: We established two areas of interest (AOI) (Fig. 8), one for the lane region directly ahead and the other for the area where the side warning messages are located. The experiment measured the driver's average duration of fixation, number of fixations, average visit duration, and number of visits to these areas of interest.

Table 2
Demonstration graphs of the four augmented conditions and comparison of the four design schemes and three types of information.

Designs	AR conditions	Information types		
		Symbol	Sign	Signal
	HMI1	No	Yes	Yes
	HMI2	Yes	Yes	Yes
	HMI3	No	Yes	No
	HMI4	Yes	Yes	No

- 2. Driving performance: Driving performance comprises the driver's decision time, acceleration at the moment of decision, and the value of deceleration. We use the acceleration at the moment of the driver's avoidance decision and the deceleration value of the vehicle within 1 s afterwards as a measure of the driver's driving performance. The decision time in the experiment begins when the patient's vehicle passes the trigger we established in the scene (for triggering the oncoming car on the right); the decision time ends when the driver executes the avoidance action. In addition, rapid acceleration and deceleration are high-risk driving behaviors that are commonly regarded as unsafe and highly correlated with the occurrence of accidents (Zhu et al., 2017). The identification of rapid acceleration and deceleration events is often performed based on fixed thresholds, and for various research purposes, 2.94 to 3.92 m/s² is a common judgment threshold (Bergasa et al., 2014; Hong et al., 2014; Toledo et al., 2008; Paefgen et al., 2012).
- 3. The after-scenario questionnaire (ASQ) (Lewis, 1991) is a commonly used task-based assessment questionnaire that evaluates system usability in terms of ease of task completion, time necessary to complete the task, and satisfaction with supporting information. The options for the scale questions range from 1 to 7, with higher scores indicating more usability, and the final ASQ score is the mean of the three values on the scale.
- 4. Workload: The driving activity load index (DALI) (Pauzié, 2008) is a driving-specific modification of the NASA-TLX. The DALI is based on the same fundamental principle as the TLX. In both tools, a scale scoring mechanism is provided for six preset parameters, followed by a weighting procedure to aggregate the six distinct scales into an overall score. The scale options range from 1 to 10, and the final score is calculated using a weighting method.
- 5. Situation Awareness: The situation awareness rating technique (SART) (Selcon and Taylor, 1990; Taylor, 2017) is a simple subjective post-test rating scale containing 10 SA dimensions: instability of situation, complexity of situation, variability of

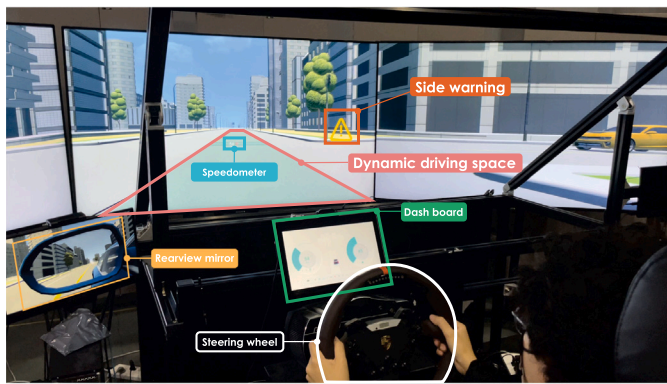


Fig. 9. Experimental setup. Note that each participant experienced all four AR-HUD designs on the simulator.

situation, arousal, concentration of attention, division of attention, spare mental capacity, information quantity, and familiarity with situation. The participants completed the scale following the experiment. Participants scored each dimension on a 7-point scale depending on their sentiments and task performance (1 = low, 7 = high). Finally, these scores were merged to assess the participants' SA.

4.2. Experiment

4.2.1. Participants

We recruited 20 participants with valid driver's licenses and normal or corrected vision through online platforms. All participants were briefed about the context of our study, both during the recruitment process and after they arrived at our experimental site. The average age of the participants was 29.7 years ($SD = 6.36$), and their average driving experience was 6.4 years ($SD = 6.20$). All participants were from Guangzhou, Guangdong Province, China, and included students, white-collar workers, and blue-collar workers. Each participant signed an informed consent form before participating in the experiment, and the average duration of their participation was approximately 30 min. They were provided with detailed information about the experiment and were compensated for their time and participation.

4.2.2. Apparatus

The experiment was conducted in a virtual driving simulation environment in our laboratory. The experimental environment consists of a fixed-base driving simulator, a primary driving system, and an experimental monitoring system. The fixed-base driving simulator and primary driving system are located in the experimental area's center, while the experimental monitoring system was located on the left side of the driving simulator, parallel to the front of the simulator. The experimental setting was encircled by soundproof panels to limit outside disturbance during the experiment.

In this experiment, a virtual driving scenario was created using Unity² software and exhibited on three 55-inch monitors with a resolution of 3840×2160 . The Tobii Glasses 2 Head Unit (serial: TG02G-010117936884) was utilized to record the eye-movement data of the individuals. Using the associated Tobii Pro Lab³ software, the eye-movement data were analyzed (see Fig. 9).

4.2.3. Procedure

This experiment required participants to maintain a consistent velocity of 40 km/h in both directions on a two-lane city street. During

this period, the participants traveled through an intersection. Various components, including sounds, sidewalks, street trees, city buildings, and so on were arranged in the driving simulator setting so that the participants would have a natural and realistic driving experience. In addition, participants did not always experience exactly the same situations, as our scenario settings have a certain degree of randomness. The vehicles driven by the participants may encounter right-turning vehicles from the right side at the intersection, or the right-turning vehicles may have already passed before the participants reached the intersection. Moreover, if the participants do not encounter right-turning vehicles during their first pass at the intersection, they will definitely encounter this dangerous situation at the next intersection. We emphasized with each participant that we expected them to decelerate to avoid sudden dangerous situations with the assistance of the information presented via the AR-HUD (WSD) interface.

Each participant was fitted with the eye-tracking device at the beginning of the experiment, and the device was calibrated using single-point calibration mode. Adjusting the height of the nose pads of the device facilitated a more pleasant wearing experience for the individual. During the experiment, the experimenter observed the participant's perspective on a Windows Surface tablet equipped with the eye-tracking recording software.

Prior to the start of the experiment, all participants were given the opportunity to experience our simulated driving simulator in a virtual driving scenario similar to the formal experiment. Participants must be somewhat familiar with using the driving simulator to manage the vehicle in a virtual setting, including speed control and directional control, because operating a driving simulator differs from driving a real vehicle in numerous ways. They were therefore also familiarized with the visual representation of information in AR-HUD through the use of interface designs from existing studies. Each participant was provided with an explanation of the graphical information variations in our designed interface, ensuring that they had a base level of familiarity with our AR-HUD interface.

After familiarization was completed, the participants began completing the experimentally determined driving task, utilizing various AR-HUD interfaces (see Fig. 7). The Latin square design was applied to counterbalance the order of interface design schemes. When a participant completed a driving activity utilizing one of the AR-HUD interfaces, the subject was asked to complete the ASQ, SART, and DALI scales.

5. Results

In this experiment, we performed data testing using SPSS 26.0 software. GraphPad Prism was utilized to visually represent the results. The measured values of each indicator were not normally distributed, as indicated by the Shapiro-Wilk test (Shapiro and Wilk, 1965). To evaluate the four AR-HUD interface scenarios, a nonparametric test based on the Friedman test (Friedman, 1937) was employed. After finding significant differences between the samples, post hoc multiple comparisons were performed using the Bonferroni-Dunn test. Furthermore, all results were adjusted for multiple comparisons using the Bonferroni correction during data analysis. The median (Mdn) with the first quartile (Q1) and the third quartile (Q3) of each variable was presented as Mdn = median (Q1, Q3).

5.1. Driving performance

A total of 20 drivers participated in this experiment. Each driver experienced all four AR-HUD design schemes. Due to a program bug, one subject's data was not recorded completely, so we excluded this participant's data.

The results of the participants' decision times when utilizing the four AR-HUD interfaces are depicted in Fig. 10(a). The shorter the decision time, the more effective the interface design solution. The Friedman

² <https://unity.cn/>.

³ <https://www.tobii.cn/>.

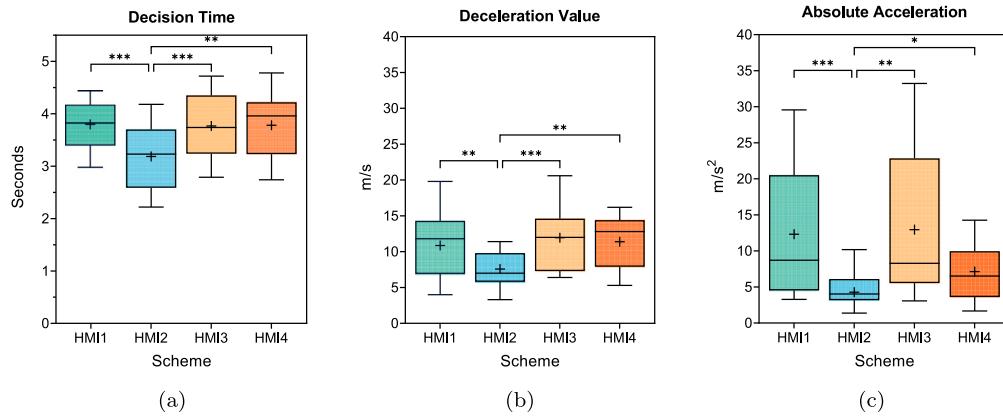


Fig. 10. Comparison of driving performance by HMI. (a) decision time, (b) acceleration at the moment of decision, and (c) deceleration. * $p < 0.05$; ** $p < 0.01$. Note that HMI2 (DSI, LW) performed the best in all three dependent variables measuring driving performance.

test revealed a statistically significant difference ($p = 0.009$) between the effects of the four HMIs on the participants' decision time. Post hoc multiple comparison analysis (Bonferroni-Dunn test) revealed that the participants' decision time was significantly shorter with HMI2 (DSI, LW) (Mdn = 3.23 (2.59, 3.70)) than with HMI1 (DSI, no LW) (Mdn = 3.82 (3.39, 4.17), $p = 0.0031$), HMI3 (QDSI, no LW) (Mdn = 3.74 (3.24, 4.35), $p = 0.0049$), and HMI4 (QDSI, LW) (Mdn = 3.96 (3.23, 4.22), $p = 0.0028$). The decision time for HMI3 (QDSI, no LW) was slightly shorter than that for HMI4 (QDSI, LW), while the decision time for HMI1 (DSI, no LW) was slightly longer than that for HMI4 (QDSI, LW), but neither was significantly different ($p > 0.05$).

Fig. 10(b) depicts the acceleration of the vehicle at the moment of participants' decision when utilizing the four AR-HUD interfaces. The smaller the absolute acceleration values, the gentler the deceleration behavior, and the more closely the absolute acceleration values approach $0.3\text{--}0.4\text{ g}$ ($g = 9.80\text{ m/s}^2$), the more efficient the interface design solution. The Friedman test revealed that the four HMIs had statistically varied effects on the acceleration of individuals at the time of decision ($p = 0.002$). Post hoc multiple comparison analysis revealed that the absolute acceleration of participants using HMI2 (DSI, LW) (Mdn = 4.03 (3.17, 6.10)) was the closest to the tolerable deceleration threshold and was significantly different from HMI3 (QDSI, no LW) (Mdn = 8.28 (5.54, 22.84), $p = 0.003$) and HMI1 (DSI, no LW) (Mdn = 8.71 (4.53, 20.51), $p = 0.003$). HMI2 (DSI, LW) was marginally superior to HMI4 (QDSI, LW) (Mdn = 6.52 (3.62, 9.96)), but not significantly ($p = 0.435$). The acceleration of HMI3 (QDSI, no LW) was less than that of HMI1 (DSI, no LW), and the acceleration of HMI1 (DSI, no LW) was less than that of HMI4 (QDSI, LW), although neither difference was statistically significant ($p > 0.05$).

Fig. 10(c) shows the results of deceleration when using the four AR-HUD interfaces. On the premise of no collision, the smaller deceleration value proves the milder the deceleration behavior and the more effective the interface design scheme. The Friedman test revealed a statistically significant difference ($p = 0.002$) in the effects of the four HMIs on deceleration. The participants' deceleration with HMI2 (DSI, LW) (Mdn = 7.00 (5.80, 9.80)) was the smallest and significantly different from HMI3 (QDSI, no LW) (Mdn = 12.00 (7.30, 14.60), $p = 0.003$) and HMI4 (QDSI, LW) (Mdn = 12.80 (7.90, 14.40), $p = 0.008$). HMI2 (DSI, LW) was marginally better than HMI1 (DSI, no LW) (Mdn = 11.80 (6.90, 14.30)), but not significantly ($p = 0.061$). The deceleration of HMI4 (QDSI, LW) was less than that of HMI3 (QDSI, no LW) and less than that of HMI1 (DSI, no LW), although neither difference was statistically significant ($p > 0.05$).

5.2. Subjective scales

Fig. 11(a) displays the results of the SART scale. Higher SART scores indicate a participant with better SA and interface solutions.

The Friedman test confirmed that the four AR-HUD interfaces had significantly distinct effects on the SA of the subjects ($p = 0.001$). Dunn's multiple comparison analysis revealed that HMI2 (DSI, LW) had the highest median SART score (Mdn = 27.50 (24.00, 31.75)) compared to both HMI1 (DSI, no LW) (Mdn = 21.00 (15.00, 26.00), $p = 0.001$) and HMI3 (QDSI, no LW) (Mdn = 22.00 (14.25, 27.00)) significantly ($p = 0.014$). The difference between the median SART scores of HMI2 (DSI, LW) and HMI4 was not statistically significant ($p = 0.402$). The difference between the SART scores of HMI4 (QDSI, LW) (Mdn = 23.50 (21.00, 27.00)) and HMI3 (QDSI, no LW) and HMI1 (DSI, no LW) was not statistically significant ($p > 0.05$). The median SART score of HMI3 (QDSI, no LW) was slightly higher than that of HMI1 (DSI, no LW), again not significant ($p > 0.05$).

Fig. 11(b) displays the results of the ASQ scale scores. Higher ASQ scores suggest a more usable solution. The Friedman test showed that the four AR-HUD interfaces had significantly distinct effects on ASQ ratings ($p = 0.0001$). Post hoc multiple comparisons revealed that HMI2 (DSI, LW) (Mdn = 6.00 (5.42, 6.68)) had the highest ASQ score when compared to HMI1 (DSI, no LW) (Mdn = 4.833 (4.33, 5.33), $p = 0.001$), HMI3 (QDSI, no LW) (Mdn = 4.833 (4.08, 5.00), $p = 0.001$), and HMI4 (QDSI, LW) (Mdn = 5.333 (4.67, 6.00), $p = 0.006$). The difference between the ASQ scores of HMI4 and HMI1 (DSI, no LW) was not statistically significant ($p = 0.345$). The difference between the ASQ scores of HMI3 (QDSI, no LW) and HMI1 (DSI, no LW) was again not statistically significant ($p = 1.000$).

Fig. 11(c) shows the results of the DALI scale. The lower the DALI score, the lighter the participant's cognitive workload and the superior the interface solution. The Friedman test indicated that the impacts of the four AR-HUD interfaces on the DALI scores were significantly different ($p = 0.0002$). Post hoc multiple comparison analysis revealed that HMI2 (DSI, LW) had the lowest DALI score (Mdn = 48.75 (35.21, 64.79)) and this difference was statistically significant ($p < 0.05$) when compared to both HMI3 (QDSI, no LW) (Mdn = 89.58 (47.08, 115.20), $p = 0.01$) and HMI4 (QDSI, LW) (Mdn = 76.25 (50.21, 95.62), $p = 0.035$). The difference between the DALI scores of HMI1 (DSI, no LW) (Mdn = 69.58 (42.29, 116.70)) and HMI3 (QDSI, no LW) was not statistically significant ($p = 0.482$). The difference between the DALI scores of HMI1 (DSI, no LW) and HMI4 (QDSI, LW) was not statistically significant ($p > 0.05$).

5.3. Eye-tracking data

Eye-tracking data can reflect the attention distribution of drivers in the driving process to some extent. On the premise that most accidents can be effectively avoided, the participants' average duration of fixation, number of fixations, and average visit duration for the front AOI are low, indicating that the information in the front AOI does

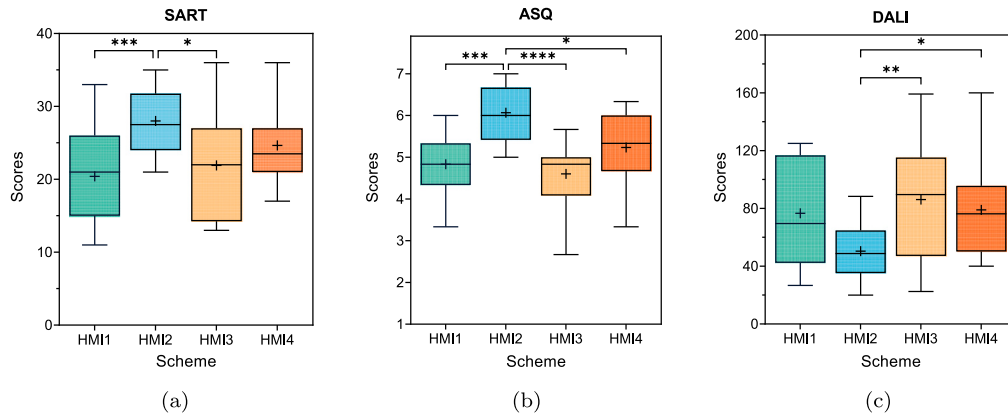


Fig. 11. Comparison of subjective scales by HMI. (a) SART score of SA, (b) ASQ score of usability, and (c) DALI score of cognitive workload. * $p < 0.05$; ** $p < 0.01$. Note that HMI2 (DSI, LW) performed the best in all three subjective scales.

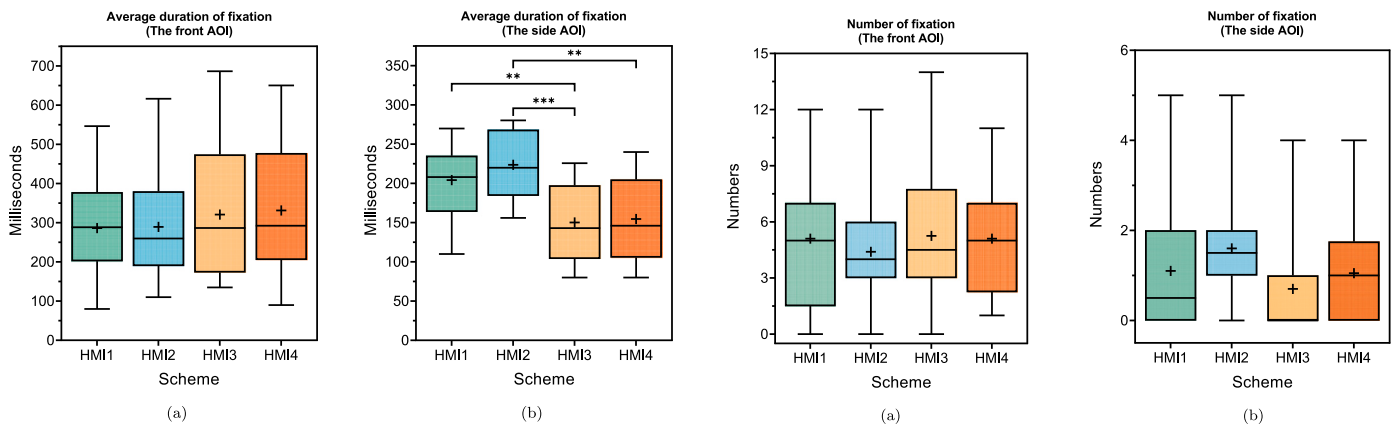


Fig. 12. Comparison of fixation duration (in ms) on both AOIs by HMI. (a) fixation duration on the front AOI, and (b) fixation duration on the side AOI. * $p < 0.05$; ** $p < 0.01$.

not distract the driver's attention excessively and can assist the driver in detecting the presence of potential hazards. As for the side AOI, the high values of these indicators mentioned above indicates that the interface solution can—to some extent—effectively guide the driver's attention to the side hazards, which is helpful to compensate for the driver's usual (unassisted) lack of side attention.

Fig. 12 displays the results for the fixation duration for each of the two AOIs (in milliseconds). The Friedman test revealed that the influence of the four AR-HUD interfaces on the fixation time of the front AOI was not significantly different ($p > 0.05$), but the effect on the fixation duration of the side AOI was significantly different ($p = 0.003$). According to the statistical description, the fixation time on the front AOI was shortest for HMI2 (DSI, LW) (Mdn = 259.80 (190.00, 379.80)). The fixation time on the side AOI was greater for HMI1 (DSI, no LW) (Mdn = 208.00 (163.80, 235.10)) than for HMI3 (QDSI, no LW) (Mdn = 143.00 (104.00, 197.40), $p = 0.009$) with significant and HMI4 (QDSI, LW) (Mdn = 146.00 (105.60, 204.90)), but less than for HMI2 (DSI, LW) (Mdn = 219.90 (184.20, 268.20)). The fixation time on the side AOI was greater for HMI2 (DSI, LW) compared to HMI3 (QDSI, no LW) with significant ($p = 0.003$) and also significant ($p = 0.009$) compared to HMI4 (QDSI, LW). The gaze time for the side AOI in HMI3 (QDSI, no LW) was lower than in HMI4 (QDSI, LW).

The results for the number of AOI fixation points are displayed in Fig. 13. The Friedman test demonstrated that the number of AOI fixation points across the various AR-HUD interfaces did not differ significantly ($p > 0.05$). HMI2 (DSI, LW) (Mdn = 4.0 (3.0, 6.0)) exhibited the lowest number of fixation points for AOI on the road ahead,

Fig. 13. Comparison of number of fixations by HMI. (a) number of fixations on front AOI, (b) number of fixations on side AOI. * $p < 0.05$; ** $p < 0.01$.

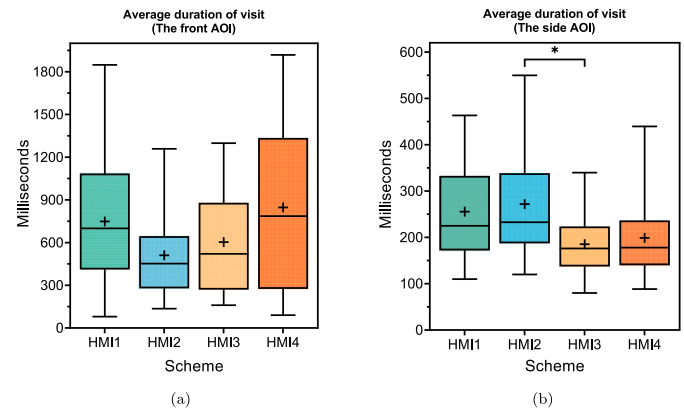


Fig. 14. Comparison of average visit duration (in ms) of the two AOIs by HMI. (a) average visit duration for front AOI, (b) average visit duration for side AOI. * $p < 0.05$; ** $p < 0.01$.

followed by HMI1 (DSI, no LW) (Mdn = 5.0 (1.5, 7.0)), HMI3 (QDSI, no LW) (Mdn = 4.5 (3.0, 7.75)), and HMI4 (QDSI, LW) (Mdn = 5.0 (2.25, 7.0)). HMI2 (DSI, LW) (Mdn = 1.5 (1.0, 2.0)) had the largest number of fixation points for the side AOI, followed by HMI1 (DSI, no LW) (Mdn = 0.5 (0, 2.0)) and HMI4 (QDSI, LW) (Mdn = 1.0 (0, 1.75)), which were both more than HMI3 (QDSI, no LW) (Mdn = 0 (0, 1.0)).

Fig. 14 displays the results of the visit duration for two AOIs (in milliseconds). The Friedman test revealed that the duration for

participants to visit the front AOI was not statistically different across the AR-HUD interfaces ($p > 0.05$), but the time for participants to visit the side AOI was significant ($p = 0.026$). According to the statistical description, the visit duration of the front AOI was longer for HMI4 (QDSI, LW) (Mdn = 786.2 (276.5, 1334.0)) than for HMI1 (DSI, no LW) (Mdn = 699.6 (413.1, 1084.0)) and HMI3 (QDSI, no LW) (Mdn = 520.8 (271.7, 878.2)), while the visit time was longer for HMI3 (QDSI, no LW) than for HMI1 (DSI, no LW). Post hoc multiple comparison analysis revealed that the visit duration of the side AOI was significantly longer ($p = 0.029$) for HMI2 (Mdn = 232.8 (187.7, 337.9)) than for HMI3 (QDSI, no LW) (Mdn = 176.1 (137.7, 223.1)). The visit duration to the side AOI when drivers used HMI4 (QDSI, LW) (Mdn = 177.9 (140.5, 236.1)) was longer than HMI3 (QDSI, no LW), but less than HMI1 (DSI, no LW).

6. Discussion

Based on the SRK taxonomy and SA theory, this paper proposes a DSID framework for AR-HUD interfaces in hazardous intersection scenarios and presents four design interfaces. The research first addresses what SA information needs to be shared with the human driver through the interface in the hazard scenario (RQ1) and what method should be used to design a DSI that meets drivers' SA needs (RQ2). In the experimental study, we compared four AR-HUD interfaces with different forms of dynamic information and varying levels of informational content to determine the impacts of different forms of dynamic information design on drivers' behavior (RQ3).

From our analyses of the driving performance data and subjective rating scores, it was clear that HMI2 with DSID and lateral warnings provided drivers with the highest level of SA and usability, and the lowest workload; HMI2 also resulted in the best driving performance among the four interfaces. This finding supports H1, indicating that well-designed DSID can enhance drivers' SA and improve their driving performance.

HMI3 (QDSI, no LW) and HMI4 (QDSI, LW) both adopted quasi-DSID, but with a lateral warning, HMI4 was slightly better than HMI3. Although HMI1 (DSI, no LW) also adopted DSID, its lack of lateral warning made it inferior to HMI2 (DSI, LW), supporting the hypothesis that DSID can reduce driver's reaction time (H2). From these experimental results, we found that AR-HUD interfaces combining warnings with DSID were most effective at enhancing drivers' driving performance and SA in dangerous scenarios.

6.1. Warning

For the purpose of safe driving, drivers are required to slow down and check for any oncoming vehicles or pedestrians when approaching an intersection. Once it is confirmed to be safe, the driver can proceed through the intersection. According to traffic regulations in China, vehicles making turns should yield to vehicles going straight. Therefore, most drivers rely on their knowledge of this regulation to prepare for potential hazards when approaching an intersection. Although this means that drivers are utilizing a knowledge-based processing mode to handle information, which is slower than a skill-based processing mode, the early appearance of warnings via an AR-HUD can help drivers prepare psychologically for the danger.

Comparative studies on symbols and alphanumeric data within HUDs have indicated that human recognition of symbols is significantly faster than text recognition (Shekhar et al., 1991). Despite interrupting the continuous safe state and temporarily releasing drivers from the control loop of dynamic driving tasks (Kaber and Endsley, 1997; Michon, 1985; Ranney, 1994), a concise 2D warning symbol effectively supports the perception stage of drivers' SA, helping drivers quickly perceive danger.

In our design, lateral warnings are implemented using the conformal design approach, presented in a world-fixed manner. Conformal design

has been demonstrated to have positive effects on directing drivers' attention and performance (Kim and Gabbard, 2022; Smith et al., 2021). In our HMIs, warnings appear before the driver enters an intersection, at a point when the driver has not yet encountered a vehicle rapidly approaching from the side (i.e., before entering a stimulus-driven task scenario). At this stage, precise attention to the right-front intersection behavior is not required, and the use of world-fixed cues can quickly guide the driver's attention to potential hazards (Merenda et al., 2018; Smith et al., 2021). Furthermore, warnings located in real space allow the driver to comprehend the information and its position without the need for additional spatial mapping processes, such as coordinating transformations between flat and spatial coordinate systems (Tönnis et al., 2009). Additionally, the use of simple graphics during the driving process can rapidly capture spatial attention, enabling the driver to selectively process visual information by prioritizing a specific area within their field of view (Karatas et al., 2020). Our experimental findings on warning information are consistent with previous research that simple graphics easily convey information to drivers, enhancing perception (Merenda et al., 2018).

Participants' statistical eye-tracking data showed that the average fixation and visit durations for the lateral warning information in HMI2 (DSI, LW) were significantly longer than in the other three HMIs. Research suggests that AR information may not actually improve drivers' perception of real-world maneuver cues, but instead influences the allocation of visual attention by increasing fixations and reducing task specificity (Eyraud et al., 2015). Moreover, longer fixations or dwell times do not necessarily indicate effective attention guidance; longer fixations may indicate that the driver is stuck in an attention tunnel or does not comprehend the graphical information. In our demonstration scenario, the lateral warning appears before the approach of the side vehicle, and its disappearance aligns with the start of color change in the DSI on the forward road, reducing the likelihood of attention tunneling. Additionally, due to the limited time available for drivers to respond to danger, there were no excessively long fixation or visitation times. Future research should further investigate the impact of lateral warnings on drivers' attention in real driving scenarios.

6.2. Dynamic spatial information

Our designed DSI is attached to the road ahead, and its content changes with the driver's behavior and situation, visualizing the relative motion positions and states between vehicles, which may help drivers establish accurate mental models (Kim and Gabbard, 2022). The dynamic changes in semantics, color, and graphic shape of the DSI conform to human behavior classification and the definition of signals at the skill-based level of the SRK taxonomy. DSI consists of an expanding red area that is independent of vehicle speed and distance traveled. As observed by Karatas et al. (2020), when the AR-HUD interface explicitly and dynamically presents hazardous objects rather than relying on metaphors or static indicators, drivers are able to identify targets more rapidly. This animation provides crucial spatial cues about the hazard and also offers predictions about the hazard's future development (Merenda et al., 2018).

According to our eye-tracking data, the average fixation time, number of fixations, and average visit duration of the participants using HMI2 (DSI, LW) are slightly less than the other three HMIs. To some extent, this represents the spatiotemporal variables of the dynamic spatial structure, which can be perceived by humans as continuous variables and thus activate drivers' behavior based on skills. DSI may also shorten the time for the driver to understand spatial information by supporting the driver's understanding of the spatiotemporal structure of the scene. Furthermore, drivers may not have to rely on high-level knowledge, experience, and reasoning to take action, thus potentially preserving more attentional resources and cognitive capacity for other elements in the environment (Bennett and Flach, 2011; Merenda et al., 2018).

DSI assists drivers' decision-making rather than replacing it, and the benefit is that it does not cause significant conflicts between interface information and drivers' mental models and SA when processing dangerous situations. In their study on the effects of the animation and boundary design of an AR-HUD's visual elements on driver attention and driving performance, Wang et al. (2023) et al. concluded that animation and boundary design have a positive impact on visual detection time. Additionally, with non-animated warnings, boundaries also positively affect brake reaction time. These experimental results, as well as the advantages of our HMI2 (DSI, LW) in average decision-making time and driving behavior mildness, provide support for this point. Dynamic cues have also been demonstrated in existing studies to reduce drivers' reaction times (Eyraud et al., 2015). Our DSID is different from previous designs that only had dynamic changes in color, semantics, or external shape in some aspects, and uses different visualization methods for other vehicle behaviors than were employed in previous designs.

In summary, the simple visual appearance of the lateral warning does not compete for excessive attentional resources with real-world objects, thereby facilitating drivers' reactions to danger in emergency situations. DSI, compared to traditional screen-fixed or world-fixed conformal cues, utilizes a world-animated approach that incorporates both directional and motion-based cues. This approach supports accurate spatial judgments to a certain extent, activating drivers' automatic sensor-motion mode and enabling them to decelerate and maneuver appropriately with a lower cognitive load when facing hazardous situations.

However, using AR graphics designed solely with the world-animated approach did not lead to the best driver performance, possibly because the drivers were required to come to a sudden stop, while the AR graphics continued to display ongoing animations. To some extent, this may lead drivers to focus solely on AR graphics without paying attention to actual road conditions (Smith et al., 2020). However, the early appearance of simple world-fixed AR graphics, as opposed to animated graphics, is more readily accepted by drivers (Smith et al., 2021). This aids drivers in perceiving danger, subsequently resulting in better outcomes for the DSI in the subsequent action phase, facilitating smoother and more effortless vehicle control.

Most of the eye-tracking data from our four interface designs did not yield significant results. This could be due to the quick redirection of participants' attention to the traffic participants displayed on the interface, and the simplicity and comprehensibility of the design elements in all four designs, which did not require drivers to allocate significant cognitive processing capacity to understand any complex or obscure information. It is possible that the task difficulty in this scenario was not high enough to cause significant differences in drivers' allocation of attentional resources across the four AR-HUD (WSD) interfaces. This calls for more precise and effective methods of measuring attention or adjustments to the urgency level of the scenario to make the changes in attentional demands more easily measurable.

7. Limitations

This study has several limitations. First, in terms of display medium, we did not compare the differences in drivers' responses to hazardous situations between the presence and absence of AR-HUD (WSD). Second, in terms of interface design, our designs were tailored specifically to the traffic rules and scenarios in China, without considering potential variations in traffic rules and scenarios in other countries and regions.

In addition, we did not thoroughly explore the effects of different expression modes in conveying AR-HUD (WSD) information on drivers' handling of hazards. Among the four interface designs used in our experiment, only world-fixed and world-animated presentations were investigated, and other expression approaches were not explored. Additionally, despite an a priori power analysis suggesting a sample size of 20 participants, our sample size was still relatively small. Moreover, we could not control for the potential influence of different drivers'

driving styles on their driving behavior when faced with hazards. Furthermore, we could not ensure that each participant had the same level of familiarity with our interface designs, making it difficult to draw clear conclusions about the impact of increasing familiarity with the interface designs on driving behavior. And the use of driving simulators and virtual driving scenarios may not fully reflect the extent to which our AR-HUD (WSD) interfaces would elicit similar responses and behaviors in the real world.

In terms of dependent variable measurement, we lacked more precise measures and analysis methods for participants' attention allocation, and the complexity and difficulty of the experimental scenarios may have influenced attention allocation and workload, resulting in limited significant findings in the eye-tracking data. Furthermore, it should be noted that the variety and intricacy of the hazardous scenarios incorporated in this study were somewhat constrained. Prior research has already indicated that the complexity of driving tasks can significantly influence driving performance when using HUDs (Smith et al., 2016). Factors such as the speed, direction, and number of approaching hazards, as well as the presence of different types of traffic participants (e.g., bicycles, pedestrians), may influence drivers' decision-making and actions when dealing with hazardous situations.

In addition to the aforementioned limitations, our proposed DSID framework also had certain constraints. Specifically, our framework only guided designers on how to design the dynamic effects of visual information but did not provide design recommendations regarding human-factors aspects, such as the color, opacity, and layout of elements. Exploring and researching these aspects will be a key focus in future work.

8. Conclusion and future work

This study investigated the impacts of AR-HUD interfaces on drivers' SA and driving performance, based on the SRK taxonomy and SA theory. In terms of theoretical contributions, we proposed a DSID framework to guide the design of AR-HUD interfaces. In our experiments, the decision time (reaction time) of drivers using HMI2 (DSI, LW) was shorter than that of drivers using our other three schemes. Moreover, drivers' driving behaviors were gentler when using HMI2 (DSI, LW), as evidenced by the acceleration being closer to the safe threshold and the deceleration being smaller. At the same time, HMI2 (DSI, LW) reduced the cognitive workload of drivers in responding to hazards, improving their user experience, as reflected in the higher average scores on the subjective scales corresponding to the use of HMI2 (DSI, LW). In addition, the statistical analyses of eye-tracking data showed that the higher average fixation duration, number of fixations, and visit duration of drivers on the side AOI using HMI2 (DSI, LW) indicated to some extent that HMI2 (DSI, LW) better directed drivers' attention to lateral hazards.

The HMI2 (DSI, LW) interface displays symbol-type side warnings, sign-type speedometers, and signal-type dynamic driving space information. Compared with other schemes, using these three types of information in HMI2 (DSI, LW) provided appropriate support for drivers' behaviors at all three levels of the SRK taxonomy in dangerous scenarios. These results also support our conclusion that DSID has the potential to help enhance driver SA in hazardous situations and improve driving safety.

In terms of practical implications, we believe that the DSID framework proposed in this study can be generalized to AR-HUD interface design in many traffic scenarios. For example, in other complex scenarios, such as left turns at intersections, AR-HUD interfaces with DSI can facilitate drivers using appropriate skill-based behaviors to deal with the actions of other traffic participants. Moreover, such interface design would not only be effective under manual driving conditions, but also at higher levels of automated driving where human and intelligent systems interact and the SA needs of the human-vehicle team change. In addition to further theoretical and methodological optimization

studies, comparative studies on different dimensions of the interface need to be explored. For example, the effects of the number of colors and the animation of the color changes on the driver's cognition and driving behavior can be investigated. It is also worth exploring the effects of different interface information layouts or representations on drivers in more complex hazardous scenarios. These issues need to be further explored in the future, and the SRK taxonomy and SA theory will continue to apply to HMI design.

CRedit authorship contribution statement

Jianmin Wang: Supervision, Writing – review & editing. **Jingyan Yang:** Data curation, Writing – original draft, Formal analysis. **Qianwen Fu:** Visualization, Investigation. **Jie Zhang:** Methodology, Supervision. **Jun Zhang:** Conceptualization, Methodology, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

This research was supported by the Fundamental Research Funds for the Central Universities; the High-Level Foreign Expert Project (G2022133031L, G2023133041L); Shenzhen Science and Technology Program (GJHZ20220913142401002).

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